

Jiaxiang Liu

ARTIFICIAL INTELLIGENCE RESEARCHER

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Summary

Jiaxiang Liu is a Postdoctoral Researcher at GDIIST, advised by Prof. Luping Shi (Tsinghua). He received his Ph.D. from Zhejiang University in 2025 under Prof. Zuozhu Liu, and was previously a Research Intern at A*STAR Singapore with Prof. Joey Tianyi Zhou and Dr. Jiawei Du. His research focuses on **trustworthy multimodal AI for medicine**, building reliable medical vision-language models via *multi-expert chain-of-thought reasoning* and *adversarial robustness*, extending to brain decoding.

He has authored **35 peer-reviewed papers** — **10 independent first-author**, **8 co-first**, **3 corresponding**, and 14 co-author — including Cell Press *Patterns* (1, Cell Press journal, IF 10.8), *Nature Communications* (1, Nature Portfolio journal, IF 18.1), ICML (1), KDD (1), NeurIPS (1), ECCV (1), ACM MM (1), AAAI (3), EMNLP (2), MICCAI (2), ACL (1), IEEE TIP (1), TCSVT (2, IF 11.1), TMM, TIM and TETCI. His **Cell Press *Patterns*** paper presented the first clinically applicable pipeline for 3D fusion and *crown-root-bone* reconstruction, featured by Cell Press; two issued patents (IP Australia + Chinese). **H-index 16**, **i10-index 22** (Google Scholar).

Honours include the **National Postgraduate Scholarship**, First Prize for Outstanding Practical Achievements at Zhejiang University, Excellent Practical Achievements of Zhejiang Province, Jiangsu Jicui Postgraduate Scholarship (4 yrs), and Ni Jihong Prize; participated in **multiple NSF and National Key R&D grants**. Reviewer for ICML (silver, 2026), NeurIPS, ICLR, KDD, CVPR, ECCV, MICCAI, AAAI, IEEE TCSVT/TIM.

Education

Zhejiang University

PH.D. IN ARTIFICIAL INTELLIGENCE, COLLEGE OF COMPUTER SCIENCE AND TECHNOLOGY

Hangzhou, China

2021 - 2025

- Supervisor: Prof. Zuozhu Liu.

Jiangxi University of Finance and Economics

M.S. IN COMPUTER TECHNOLOGY

Nanchang, China

2018 - 2021

- Supervisor: Prof. Yong Yang.

Jiangxi Normal University

B.S. IN INTERNET OF THINGS, SCHOOL OF COMPUTER INFORMATION ENGINEERING

Nanchang, China

2014 - 2018

Work Experience

Guangdong Institute of Intelligence Science and Technology (GDIIST)

POSTDOCTORAL RESEARCHER

Hengqin, Zhuhai, China

2025 - PRESENT

- Supervisor: Prof. Luping Shi (Tsinghua University).

Agency for Science, Technology and Research (A*STAR)

RESEARCH INTERN

Singapore

Sep. 2023 - Sep. 2024

- Supervisors: Prof. Joey Tianyi Zhou and Dr. Jiawei Du.

Publications

First Author (Independent) × 10

(1 Cell Press journal, 6 CCF-A / TH-CPL-A conferences, 2 JCR Q1 journals, 1 flagship medical conference)

Co-First Author × 8 | **Corresponding Author × 3** | **Co-Author × 14** (incl. 1 Nature Communications)

Symbols: # co-first author, * corresponding author.

SELECTED FIRST-AUTHOR CONFERENCE PAPERS

BioMedVR: Confusion-Aware Mixture-of-Prompt Experts for Biomedical Visual Reprogramming [PDF]

ECCV 2026

JIAXIANG LIU, TIANXIANG HU, JUWEI GUAN, YUJIE WU, YUSONG WANG, YAO MU, ZUOZHU LIU, MINGKUN XU

2026

CCF-B / TH-CPL-A. First input-space dual-expert visual reprogramming for biomedical VLMs.

Self-Calibrated Consistency can Fight Back for Adversarial Robustness in Vision-Language Models [PDF] JIAXIANG LIU , JIAWEI DU, XIAO LIU, SHANGYANG LI, SONGCHEN MA, CHANGSHUO WANG, PRAYAG TIWARI, MINGKUN XU CCF-A / TH-CPL-A. arXiv:2510.22785; code at github.com/JXLiu-AI/SCC	ICML 2026 2026
MindAdapter: Few-Shot Parameter-Efficient Residual Calibration of Cross-Subject Brain-to-Visual Decoding Models [PDF] JIAXIANG LIU , JIAWEI DU, XUPENG CHEN, GUOQI LI, JIANG CAI, SIMON FONG, MINGKUN XU CCF-A / TH-CPL-A. Project page: jxliu-ai.github.io/MindAdapter	KDD 2026 (AI4Sciences track) 2026
KPL: Training-Free Medical Knowledge Mining of Vision-Language Models [PDF] JIAXIANG LIU , TIANXIANG HU, JIAWEI DU, RUIYUAN ZHANG, JOEY TIANYI ZHOU, ZUOZHU LIU CCF-A. Code: github.com/JXLiu-AI/KPL	AAAI 2025 (Oral) 2025
VPL: Visual Proxy Learning Framework for Zero-Shot Medical Image Diagnosis [PDF] JIAXIANG LIU , TIANXIANG HU, HUIMIN XIONG, JIAWEI DU, YANG FENG, JIAN WU, JOEY TIANYI ZHOU, ZUOZHU LIU TH-CPL / CAAI A	EMNLP 2024 2024
MedCoT: Medical Chain of Thought via Hierarchical Expert [PDF] JIAXIANG LIU , YUAN WANG, JIAWEI DU, JOEY TIANYI ZHOU, ZUOZHU LIU TH-CPL / CAAI A. Code: github.com/JXLiu-AI/MedCoT	EMNLP 2024 (Oral) 2024
ToothSegNet: Image Degradation Meets Tooth Segmentation in CBCT Images [PDF] JIAXIANG LIU , TIANXIANG HU, YANG FENG, WANGHUI DING, ZUOZHU LIU Flagship medical conference. Code: github.com/JXLiu-AI/Tooth-UNet	IEEE ISBI 2022
FIRST-AUTHOR JOURNAL PAPERS	
Deep learning-enabled 3D multimodal fusion of cone-beam CT and intraoral mesh scans for clinically applicable tooth-bone reconstruction [PDF] JIAXIANG LIU , JIN HAO, HANGZHENG LIN, WEI PAN, JIANFEI YANG, YANG FENG, GAOANG WANG, JIN LI, ZHIHE ZHAO, ZUOZHU LIU Cell Press data-science journal	Cell Press Patterns 2023
Parameter-Efficient Transfer Learning for Medical Visual Question Answering [PDF] JIAXIANG LIU , TIANXIANG HU, YAN ZHANG, YANG FENG, JIN HAO, JUNHUI LV, ZUOZHU LIU JCR Q1.	IEEE TETCI 2023
IFCoT: Insights Fusion Chain of Thought for Multimodal Reasoning in LLMs JIAXIANG LIU , JIAWEI DU, TIANXIANG HU, LINJIE TONG, YAN ZHANG, BING LI, YANG FENG, JIAN WU, JOEY TIANYI ZHOU, ZUOZHU LIU JCR Q1 / SCI Q1. IF 11.1	IEEE TCSVT (Accepted) 2026
Modest-Align: Data-Efficient Alignment for Vision-Language Models [PDF] JIAXIANG LIU , YUAN WANG, JIAWEI DU, JOEY TIANYI ZHOU, MINGKUN XU, ZUOZHU LIU JCR Q1. arXiv:2510.21606; code: github.com/JXLiu-AI/Modest-Align	Pattern Recognition (Under Review) 2025
CO-FIRST AUTHOR / CORRESPONDING AUTHOR PAPERS (SELECTED)	
3D-RAD: A Comprehensive 3D Radiology Med-VQA Dataset with Multi-Temporal Analysis and Diverse Diagnostic Tasks [PDF] XIAOTANG GAI, JIAXIANG LIU #, YU LI, ZIJIE MENG, JIAN WU, ZUOZHU LIU CCF-A / TH-CPL-A. Co-first author	NeurIPS 2026 (Dataset Track) 2026
Med-GLIP: Advancing Medical Language-Image Pre-training with Large-scale Grounded Dataset [PDF] ZIYE DENG, RUIHAN HE, JIAXIANG LIU #, YUAN WANG, ZIJIE MENG, SONGTAO JIANG, YONG XIE, ZUOZHU LIU CCF-B. Co-first author	IEEE BIBM 2025
GTCL: Graph-Text Contrastive Learning Meets Log Anomaly Detection [PDF] XIAO TAN, JIAXIANG LIU #, YUJIE WU, MINGKUN XU CCF-B. Co-first author	IEEE ICASSP 2026
Fair-MoE: Fairness-Oriented Mixture of Experts in Vision-Language [PDF] LINJIE TONG, PEIRAN WANG, JIAXIANG LIU *, ZUOZHU LIU CCF-B. Corresponding author	MICCAI 2025

Infrared and Visible Image Fusion based on Dual-kernel Side Window Filtering and S-shaped Curve Transformation

IEEE TIM

YONG YANG, WANG ZHANG, SHUYING HUANG, WEIGUO WAN, **JIAXIANG LIU**, XIANGKAI KONG
JCR Q1.

2021

ECLB: Efficient Contrastive Learning on Bi-Level for Noisy Labels

Knowledge-Based Systems (KBS)

JUWEI GUAN, **JIAXIANG LIU**, SHUYING HUANG, YONG YANG
JCR Q1 / SCI Q1.

2024

A ChatGPT-Aided Explainable Framework for Zero-Shot Medical Image Diagnosis [PDF]

ICML-IMLH Workshop

JIAXIANG LIU, TIANXIANG HU, YAN ZHANG, XIAOTANG GAI, YANG FENG, ZUOZHU LIU

2023

Patents

Multiple Decision Maps Based Infrared and Visible Image Fusion

IP Australia, No. 2020100178

YONG YANG (SUPERVISOR); **LIU, JIAXIANG**; ET AL.

2021

Image Fusion Method, Device, Equipment and Readable Storage Medium

Chinese Patent, No. 202311022450

YANG FENG; ZUOZHU LIU; **LIU, JIAXIANG**

2023

Research Projects

Multi-Expert Verified Medical Chain-of-Thought Reasoning (MedCoT framework)

Hangzhou, China

SCIENTIFIC RESEARCH FUND OF ZHEJIANG UNIVERSITY

2023 - 2025

- Hierarchical experts + Sparse MoE + feature-level voting; a 256M model outperforms LLaVA-Med (7B) by **5%+**.
- **Mentoring**: mentored 2 master students, who published 4 papers at **NeurIPS, NAACL, AAAI, MICCAI**.
- **Personal outputs**: 4 top-tier conference papers as lead/co-lead author (**MedCoT, KPL, VPL, MedThink**); awarded **First Prize** at Zhejiang University and the **Outstanding Practical Achievement Award** of Zhejiang Province.

Clinical applicable multimodal fusion algorithm research for 3D intraoral scanning and CBCT data

Hangzhou, China

SCIENTIFIC RESEARCH FUND OF ZHEJIANG UNIVERSITY

2022 - 2023

- Reduced multimodal oral data *crown-root-bone* reconstruction time from a conventional **5-7 hours** to **20-25 minutes** (~15x speed-up).
- Algorithm deployed in the **Angelalign IRS** clinical orthodontic system; results featured by **Cell Press** in an official news release.
- **Mentoring**: mentored 1 undergraduate student, who published 3 papers at **MIDL, MICCAI, BIBM**.
- **Personal outputs**: **Cell Press Patterns 2023** first-author paper; awarded **First Prize** at Zhejiang University and **Excellent Practical Achievement** of Zhejiang Province.

Infrared and visible image fusion for public safety

Nanchang, China

STUDENT RESEARCH PROJECT OF JIANGXI UNIVERSITY OF FINANCE AND ECONOMICS

2020 - 2021

- Designed multi-scale and modal feature fusion networks for IR-VIS image fusion in public-safety scenarios.
- Outputs: 4 IEEE journal / conference papers (TCSVT, TMM, TIM, ICME Oral) and 1 international patent (IP Australia).

Skills & Service

Programming	Python, MATLAB, C, HTML5, SQL.
Languages	Native Chinese; Fluent English.
Editorial service	Young Editorial Board Member, <i>Computer and Modernization</i> .
Conference reviewer	ICML (silver tier, 2026), NeurIPS, ICLR, KDD, CVPR, ECCV, MICCAI, AAAI.
Journal reviewer	IEEE TCSVT, IEEE TIM, IEEE TIP.

Honors & Awards

2024	First Prize (university level) , Outstanding Practical Achievements of Professional Degree Graduate Students, Zhejiang University	Hangzhou, China
2024	Provincial Award , Outstanding Practical Achievements of Professional Degree Graduate Students, Zhejiang Province	Zhejiang, China
2022–2025	Multiple Graduate Scholarships , Jiangsu Province Jicui Postgraduate Scholarship (4 yrs) & Ni Jihong Prize, Zhejiang University	China
2021	National Scholarship , National Scholarship for Postgraduates	China
2019–2020	National & Provincial 1st–3rd Prizes (5 total) , Huawei Cup China Post-Graduate Mathematical Modeling (16th 2nd; 17th 3rd) & Jiangxi Province Modeling (1st, 2nd) & 7th TipDM Cup Data Mining (2nd)	China
2018–2025	Excellent Graduate , Jiangxi Normal University & Zhejiang University	China
2015	Provincial First Prize , Jiangxi Province University Student Science and Technology Innovation and Skills Competition	Jiangxi, China